Robust numerical scheme for stochastic copula models

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Abstract

This study aims to develop a copula model for constructing multivariate distributions considering the dynamics of financial assets. We propose a stochastic copula model in continuous time using an Ornstein-Uhlenbeck equation as a time-dependent copula parameter. The Markov property and known transition density of the Ornstein-Uhlenbeck process allow the high-dimensional likelihood integral to be factored into a sequence of two-dimensional integrals computed via the transfer matrix method on a discretized grid. This approach yields a numerically exact evaluation of the likelihood function, avoiding the variance and bias inherent in Monte Carlo estimators. We discuss implementation details including adaptive grid refinement and the use of sparse transfer matrices for long time series. The method is extended to high-dimensional settings via C-vine pair copula constructions, where each edge copula is equipped with its own stochastic parameter. The proposed model is validated on cryptocurrency market data across four Archimedean copula families in the bivariate case and a six-dimensional C-vine copula, demonstrating consistent and significant improvements in both log-likelihood and goodness-of-fit compared to static copula models.

Keywords Stochastic copula models, Transfer matrix method, Ornstein-Uhlenbeck process, Vine copulas, Adaptive grid, Sparse matrices

1 Introduction

Modeling the joint behavior of financial asset returns requires capturing dependencies that may evolve over time. Copula models provide a flexible framework for constructing multivariate distributions by separating the modeling of marginal distributions from the dependence structure [Skl59, Nel07]. However, standard copula models with fixed parameters cannot account for the time-varying nature of dependencies observed in financial markets.

In our previous article [SS19], we presented the results of modeling risk measures based on the copula approach for an optimal portfolio on short financial time series comprising 253 observations in total. Our findings showed that the R-vine copula model has a higher predictive ability compared to elliptical copula models. However, in the previous study we did not account for the possibility of dynamic changes in the relationship between variables. While this oversight may not be critical in short time series, it becomes increasingly important when modeling long time series. It is necessary to take into account the dynamic changes in the relationships observed in the original data.

In the review [MR12] the authors distinguished between parametric, semi-parametric and non-parametric copula models that account for dynamic changes in the variable relationships. There is a brief outline of approaches.

Among the parametric models, we highlighted dynamic conditional correlation models and their further development into asymmetric dynamic conditional correlation models [KHKK23]. These models take into account the dynamics of conditional correlations through their dependence on previous shocks values (errors), adjusted for volatility. Autoregressive models and generalized autoregressive score (GAS) are based on observation-driven copula models [CKL13], where the dependence of the copula on time is described by a parametric function with an autoregressive term. Stochastic autoregressive copula models (SCAR) [HM12] extend this concept by including a parameter-driven model responsible for the change in the model over time, which is introduced using an independent stochastic process, rather than data as in previous models. Further development of the SCAR and GAS models have been proposed in [ACM16, KMM22] using D-vine-based dynamic copula models. In these models, time-varying dependence parameters of copula model is achieved by introducing time series for copula parameters based on autoregressive moving average and generalized autoregressive conditional heteroskedasticity models. In [Mba24] the author propose using a vine-copula in conjunction with the asymmetric power autoregressive conditional heteroscedasticity (APARCH) model to more accurately model the dependency between assets and account for complex dependencies and tail dynamics in the portfolio.

In the class of nonparametric models, we refer to the paper [Har10] where the authors use a quantile function (estimating time-varying quantiles) to decide on a change in the model. The review by [MR12] presents several models where parameter responsible for the change in the model over time depends on the marginal distribution through the conditional variance [vdGGW05]. The article [NV23] introduces a two-dimensional dynamic stochastic copula, built on time series of different periodicities: daily for asset prices (stock index, oil price), and monthly for factors (inflation, interest rate).

A common limitation of Monte Carlo approaches used in existing SCAR models [HM12, LR03] is the bias inherent in the log-mean-exp estimator: by Jensen's inequality, $\mathbb{E}[\log \hat{L}] \leq \log L$, with the bias decaying only as $O(1/N)$ in the number of trajectories. For long time series this bias can be substantial, requiring an impractically large number of Monte Carlo paths.

The purpose of the paper is to develop a copula model for constructing multivariate distributions that account for dynamics of financial assets. The proposed model incorporates a continuous-time stochastic process, the Ornstein-Uhlenbeck process, to capture the time-varying dependencies between assets. The key methodological contribution is the use of the *transfer matrix* approach, which exploits the Markov structure and known Gaussian transition density of the OU process to reduce the high-dimensional likelihood integral to a sequence of numerically tractable two-dimensional integrals evaluated on a grid via quadrature.

The primary objectives of this research are twofold: (1) to introduce time dynamics as a copula parameter through a stochastic differential equation in continuous time with a numerically exact likelihood evaluation via the transfer matrix method, and (2) to evaluate the effectiveness of the novel copula model using real datasets of different lengths from the cryptocurrency market.

The key contributions of this paper are as follows.

Firstly, we propose a novel stochastic copula model for continuous cases based on the Ornstein-Uhlenbeck process. Secondly, we develop a transfer matrix scheme that yields a numerically exact

evaluation of the likelihood function, avoiding the variance and bias inherent in Monte Carlo estimators used in earlier SCAR models. We discuss implementation aspects of the transfer matrix method, including adaptive grid refinement and the use of sparse matrices to handle long time series efficiently. Thirdly, we extend the model to the multivariate setting via C-vine pair copula constructions, where each edge copula is equipped with its own latent OU process and fitted via the transfer matrix approach.

The implementation of the proposed model is available as an open-source Python package at https://github.com/AANovokhatskiy/pyscarcopula.

This paper is structured as follows. Section 2 reviews various types of copula models, including Archimedean copulas, vine copula constructions, and stochastic autoregressive copulas (SCAR). The continuous SCAR model and the transfer matrix method for exact likelihood computation are developed in Section 3, including implementation details for the numerical scheme. Section 4 focuses on goodness-of-fit tests for proposed copula models. Numerical experiments with real datasets from the cryptocurrency market are presented in Section 5, including bivariate and multivariate model comparisons across multiple copula families, convergence analysis, performance benchmarks, and a quantification of the Monte Carlo bias. Finally, Section 6 concludes the paper and discusses the future work.

2 Copula models review

Consider a set of univariate random variables $X_1, ..., X_d$. In the case of financial datasets this variables could have a complex dependencies. To take them into account one could construct a multivariate distribution. One way to do this based on Sklar's theorem and copula approach [Skl59].

Let $\boldsymbol{X} = (X_1, ..., X_d)$ – d-dimensional random variable. Distribution function of $\boldsymbol{X}$ could be represented as

$$F(x_1, \ldots, x_d) = C(F_1(x_1), \ldots, F_d(x_d)), \quad (x_1, \ldots, x_d) \in \mathbb{R}^d,$$

where function C is d-dimensional function of one or more parameters called *copula*, which strict definition and properties could be found at [Nel07]. It is useful to introduce vector $\boldsymbol{U} = (U_1, ..., U_d) = (F_1(x_1), ..., F_d(x_d))$ whose components are uniform at $[0, 1]$. Hence we can write the expression of copula density

$$c(u_1, ..., u_d) = \frac{\partial^d C(u_1, ..., u_d)}{\partial u_1 \cdots \partial u_d}, \quad u_1, ..., u_d \in [0, 1]$$

and density of joint distribution

$$f(x_1, \ldots, x_d) = c(F_1(x_1), \ldots, F_d(x_d)) \prod_{i=1}^{d} f_i(x_i), (x_1, \ldots, x_d) \in \mathbb{R}^d. \tag{2.1}$$

General problem of finding parameters of joint distribution implies solving the optimization problem for whole function (2.1), but it is convenient to solve two separate problems: for copula and for marginal distributions. For copula one can use the following procedure. Let us define $U_i = F_i(X_i)$ – uniformly disturbed marginals and corresponding variables $u_i = F_i(x_i)$, $i = 1, ..., d$. Then copula density will take the form $c(u_1, ..., u_d; \theta)$. Calculation of variables u_i implies the knowledge of the distribution function, but we can estimate the function $F_i(x_i)$ by non-parametric way using it's empirical version, which is equivalent to introducing a new variables called *pseudo observations*:

$$u_{ij} = \frac{rank(x_{ij})}{n+1}, \quad i = 1, ..., d, \quad j = 1, ..., n, \tag{2.2}$$

where $rank(x_{ij})$ – element number of x_{ij} in ranked series $x_{i1}, ..., x_{in}$. Denote as $U_i = \{u_{ij}\}_{j=1}^n$ – a set of observations, $i = 1, ..., d$. Then the maximum likelihood estimation of copula parameter θ_c is (such an approach could be found in literature as Omnibus or OM estimation)

$$\theta_c = \arg\max_{\theta} \log L(U_1, ..., U_d; \theta) = \sum_{j=1}^{n} \log c(u_{1j}, ..., u_{dj}; \theta).$$

2.1 Archimedian copulas

In order to construct a multivariate distribution based on copulas one should use a some class. In the study we used Archimedian copulas.

Consider a strictly decreasing convex function $\phi(t;\theta)$ such that $\phi(1;\theta) = 0$, θ is fixed parameter. Function ϕ is called *generator* of copula. Formally the Archimedean copula is a function satisfying the following parametrization

$$C(u_1, ..., u_d) = \phi^{-1}\left(\phi(u_1) + ... + \phi(u_d)\right).$$

The following theorem [Kim74] establish necessary and sufficient conditions on generator ϕ to generate a multivariate copula.

Theorem 1 *Let ϕ be a continuous strictly decreasing function from $[0,1]$ to $[0,\infty)$ such that $\phi(0) = \infty$ and $\phi(1) = 0$, and let ϕ^{-1} denote the inverse of ϕ. Then $C(u_1, ..., u_d)$ is an d-copula for all $n \geq 2$ if and only if ϕ^{-1} is completely monotone on $[0,\infty)$, i. e. $(-1)^k \frac{d^k \phi^{-1}(t)}{dt^k} \geq 0$ for any $k \in \mathbb{N}$.*

The generators of some Archimedian copulas in explicit form are given in Table 1 (see [Nel07] for details). For given copulas the conditions of Theorem 1 holds.

Table 1: Archimedian copulas

Copula	$\phi(t;\theta)$	$\phi^{-1}(t;\theta)$	$\theta \in$
Gumbel	$(-\log t)^\theta$	$\exp\left(-t^{1/\theta}\right)$	$[1,\infty)$
Frank	$-\log\left(\frac{\exp(-\theta t)-1}{\exp(-\theta)-1}\right)$	$-\frac{1}{\theta}\log\left(1+\exp(-t)(\exp(-\theta)-1)\right)$	$(0,\infty)$
Joe	$-\log\left(1-(1-t)^\theta\right)$	$1-(1-\exp(-t))^{(1/\theta)}$	$[1,\infty)$
Clayton	$\frac{1}{\theta}\left(t^{-\theta}-1\right)$	$(1+t\theta)^{-1/\theta}$	$(0,\infty)$

In the bivariate case it is useful to introduce a rotated (or survival) copulas. The copulas rotated on 90, 180, 270 degrees respectively have the form:

$$\begin{aligned} C_{90}(u_1,u_2) &= u_2 - C(1-u_1, u_2), \\ C_{180}(u_1,u_2) &= u_1 + u_2 - C(1-u_1, 1-u_2), \\ C_{270}(u_1,u_2) &= u_1 - C(u_1, 1-u_2). \end{aligned}$$

The form of the density does not change accurate to variable substitution. For example, $c_{180}(u_1,u_2) = c(1-u_1, 1-u_2)$.

2.2 Sampling Archimedian copulas

An efficient sampling algorithm for Archimedean copulas was proposed by Marshall and Olkin [MO88]. It relies on the fact that the inverse generator $\psi(t) = \phi^{-1}(t)$, when completely monotone, is the Laplace-Stieltjes transform of some distribution function F. The algorithm proceeds as follows: (1) sample $V \sim F = \mathcal{LS}^{-1}(\psi)$; (2) sample i.i.d. $X_i \sim U[0,1]$, $i = 1,\ldots,d$; (3) return $(U_1,\ldots,U_d)$ where $U_i = \psi(-\log X_i/V)$. The required inverse Laplace-Stieltjes transforms for the copulas in Table 1 are known explicitly; see Hofert [Hof08] and Joe [Joe97] for details and numerical studies.

2.3 Vine copulas

Archimedean and elliptical copulas impose restrictive symmetry conditions on the dependence structure, which limits their applicability in dimensions $d > 2$. Vine copulas, introduced by Bedford and Cooke [BC02], offer a flexible alternative by decomposing a d-dimensional copula density into a product of $d(d-1)/2$ bivariate copulas, each of which can be chosen from a different family. This pair copula construction (PCC) was developed in detail by Aas et al. [ACFB09] and is extensively treated in Czado [Cza19].

A *regular vine* on d variables is a sequence of $d-1$ linked trees $T_1,\ldots,T_{d-1}$. Each tree T_j has $d-j$ edges, and each edge corresponds to a conditional bivariate copula. Two commonly used special cases are the *C-vine* (canonical vine), where each tree has a designated root node, and the *D-vine*, where each tree forms a path.

In a C-vine, the density decomposes as

$$c_{1:d}(\mathbf{u}) = \prod_{j=0}^{d-2} \prod_{i=1}^{d-j-1} c_{j+1,\, j+i+1\,|\,1:j}\big(u_{j+1|1:j},\ u_{j+i+1|1:j}\big)\,,$$

where $c_{a,b|S}$ denotes the density of the bivariate copula for variables a and b conditional on the set S, and $u_{a|S}$ is the conditional probability integral transform (obtained via the h-function). In tree T_0 we model the unconditional pairs $(1,2),(1,3),\ldots,(1,d)$; in tree T_1 we model the pairs $(2,3|1),(2,4|1),\ldots$, and so on. Each bivariate copula in the construction is called an *edge copula* and can be selected from a different parametric family (Gumbel, Clayton, Frank, etc.) via an information criterion such as AIC [Cza19].

The conditional pseudo-observations $u_{a|S}$ needed at higher tree levels are computed recursively via the h-function:

$$h(u_2, u_1; \theta) = \frac{\partial C(u_1, u_2; \theta)}{\partial u_1} = C(u_2|u_1; \theta). \tag{2.3}$$

Specifically, if edge $(a, b|S)$ at tree level j uses copula $C_{a,b|S}$ with parameter θ, then the pseudo-observation for the next tree level is $u_{b|S\cup\{a\}} = h(u_{b|S},\, u_{a|S};\, \theta)$.

The combination of vine copulas with the stochastic copula model proposed in this paper is discussed in Section 3.5.

2.4 Stochastic autoregressive copulas (SCAR)

Consider a single-parameter model, for example an Archimedean copula or a bivariate elliptical copula. Without loss of generality consider the bivariate case and suppose that the copula parameter is not fixed and follows the law

$$\begin{aligned}(u_{1t}, u_{2t}) &\sim c(u_{1t}, u_{2t}; \theta_t),\\ \theta_t &= \Psi(\lambda_t),\\ \lambda_t &= \alpha_1 + \alpha_2\lambda_{t-1} + \alpha_3\eta_t,\end{aligned} \tag{2.4}$$

where $\Psi : \mathbb{R} \to \Theta$ is a suitable transformation that guarantees that the copula parameter remains in its domain of definition. It is assumed that the underlying process $\{\lambda_t\}_{t=1}^T$ is latent (i.e. unobservable) and is described by a Gaussian autoregressive process of the first order $AR(1)$, where $\eta_t \overset{iid}{\sim} \mathcal{N}(0,1)$, $|\alpha_2| < 1$ to ensure stationarity of λ_t and $\alpha_3 > 0$. This approach was suggested in Liesenfeld and Richard [LR03], discussed among other time-varying models in Manner and Reznikova [MR12] and was developed in Almeida, Czado, Manner [ACM16] for vine copulas.

Since the process $\{\lambda_t\}_{t=1}^T$ is unobservable, estimation for $\alpha = (\alpha_1, \alpha_2, \alpha_3)$ cannot be obtained explicitly. The likelihood function has the form

$$L(\alpha; U_1, U_2) = \int \prod_{t=1}^{T} c\left(u_{1t}, u_{2t}; \Psi(\lambda_t)\right) p\left(\lambda_t \,|\, \lambda_{t-1}; \alpha\right) d\Lambda, \tag{2.5}$$

where $U_{1,t} = \{u_{1i}\}_{i=1}^t$, $U_{2,t} = \{u_{2i}\}_{i=1}^t$ is a set of pseudo-observations known up to time t, and $p(\lambda_t|\lambda_{t-1};\alpha)$ is the transition density. A straightforward Monte Carlo estimate of (2.5) is obtained by sampling N trajectories from the prior density p:

$$\hat{L}_N(\alpha; U_1, U_2) = \frac{1}{N}\sum_{i=1}^{N}\prod_{t=1}^{T} c\left(u_{1t}, u_{2t}; \Psi\left(\lambda_t^{(i)}\right)\right), \tag{2.6}$$

where $\lambda_t^{(i)}$ is the value of the i-th trajectory at time t. This estimator, referred to as the *p-sampler*, is unbiased for L but is very inefficient for large T: the variance of the product grows exponentially, and the log-likelihood estimate $\log \hat{L}_N$ is biased downward by Jensen's inequality.

The Monte Carlo variance can be reduced by *efficient importance sampling* (EIS) [LR03]: one introduces an auxiliary density $m(\lambda_t|\lambda_{t-1}; a_t)$ with parameters a_t chosen to minimize the variance of the importance weights. The resulting estimator, known as the *m-sampler*, takes the form

$$\hat{L}_N(\alpha; U_1, U_2) = \frac{1}{N}\sum_{i=1}^{N}\prod_{t=1}^{T} \frac{c\Big(u_{1t}, u_{2t}; \Psi\Big(\lambda_t^{(i)}\Big)\Big)\ p\left(\lambda_t^{(i)}|\lambda_{t-1}^{(i)}; \alpha\right)}{m\left(\lambda_t^{(i)}|\lambda_{t-1}^{(i)}; a_t\right)},$$

where the trajectories $\lambda_t^{(i)}$ are now sampled from the density m. While the m-sampler substantially reduces the required number of trajectories (typically $100 < N < 200$), both Monte Carlo approaches remain subject to the Jensen bias in log-likelihood and to practical difficulties in the continuous-time setting (see Section 3.3 for a deterministic alternative).

3 Continuous SCAR model

3.1 Model formulation

Suppose now that the process λ_t is a process in continuous time on the interval $t \in [0, 1]$. The continuous version of the discrete process (2.4) is the Ornstein-Uhlenbeck process

$$dx_t = (\alpha_1 + \alpha_2 x_t)\, dt + \alpha_3 dW_t, \quad t \geq 0, \tag{3.1}$$

where W_t – Wiener process. The same in canonical variables:

$$dx_t = \theta\,(\mu - x_t)\, dt + \nu dW_t, \quad t \geq 0, \tag{3.2}$$

where the parameters of the processes (3.1) and (3.2) are related as follows:

$$\mu = -\frac{\alpha_1}{\alpha_2}, \quad \theta = -\alpha_2, \quad \nu = \alpha_3.$$

Let the probability space $(\Omega, \mathcal{F}, P)$ be given. Consider on it the random process $\{x_t\}$, $t \in [0, 1]$. Let us divide the interval $[0, 1]$ by the points $0 \leq t_1 \leq t_2 \leq \ldots \leq t_T \leq 1$. Let us introduce the following probability measure:

$$\nu_{t_1,\ldots,t_T}\,(F_1 \times \ldots \times F_T) = \int\limits_{F_1 \times \ldots \times F_T} \prod_{k=1}^{T} p(x_k, t_k | x_{k-1}, t_{k-1})\, dx_1 dx_2 \ldots dx_T,$$

where $p(x, t|x', t')$ – probability density function of transition from the state (x', t') to (x, t). We denote as $\alpha = (\theta, \mu, \nu)$ the set of constant parameters of stochastic equation (3.2) in canonical form. By analogy with the expression (2.5), define the likelihood function as follows:

$$\begin{aligned} L(\alpha; U_1, U_2) &= \mathbb{E}\left(\prod_{t=1}^{T} c(u_{1t}, u_{2t}, \Psi(x_t) | U_{1,t-1}, U_{2,t-1}, x_{t-1}; \alpha)\right) = \\ &= \int\limits_{F_1 \times \ldots \times F_T} \prod_{t=1}^{T} c(u_{1t}, u_{2t}, \Psi(x_t) | U_{1,t-1}, U_{2,t-1}, x_{t-1}; \alpha)\, d\nu_{t_1,\ldots,t_T}\,(F_1 \times \ldots \times F_T)\,, \end{aligned} \tag{3.3}$$

where the partitioning of the interval $[0, 1]$ is made by T points – according to a given set of observations. Now define the stochastic copula model in the continuous case

$$\begin{aligned} (u_{1t}, u_{2t}) &\sim c(u_{1t}, u_{2t}; \theta_t), \\ \theta_t &= \Psi(x_t), \\ dx_t &= \theta\,(\mu - x_t)\, dt + \nu dW_t, \quad t \in [0, 1], \end{aligned}$$

where $\Psi(x_t)$ is a real function that translates the values of the process x_t into the interval of parameter values allowed for a given copula. In further calculations, we will use $\Psi(x) = x \tanh(x)$ for Clayton and Frank copulas, and $\Psi(x) = 1 + x \tanh(x)$ for Gumbel and Joe.

The choice of Ψ is guided by three requirements: (i) the range must match the parameter domain of the copula ($\theta > 0$ for Clayton and Frank, $\theta \geq 1$ for Gumbel and Joe); (ii) Ψ must be smooth so that the integrand $f_t(z) = c(u_{1t}, u_{2t}; \Psi(z + \mu))$ is amenable to quadrature; and (iii) Ψ should be bounded from below but grow for large $|x|$ to allow the copula parameter to vary over a wide range. The function $x \tanh(x)$ satisfies all three: it is infinitely differentiable, nonnegative with $\Psi(0) = 0$, and grows as $|x|$ for $|x| \to \infty$. Moreover, the symmetry $\Psi(x) = \Psi(-x)$ is natural in the centered coordinates $z = x - \mu$ of the OU process. We considered alternative link functions such as exp and softplus, but these grow too rapidly (or too slowly), leading to numerical instability in the copula density evaluation at extreme grid points.

3.2 Transition density of the OU process

For a continuous stochastic Ito process of the form

$$dx_t = A(x,t)dt + B(x,t)dW_t,$$

the transition density $p(x,t|x',t')$ satisfies the Fokker-Planck equation

$$\frac{\partial p(x,t)}{\partial t} = -\frac{\partial}{\partial x}\left(A(x,t)p(x,t)\right) + \frac{\partial^2}{\partial x^2}\left(\frac{B^2(x,t)}{2}p(x,t)\right). \tag{3.4}$$

For the OU process (3.2), the solution of (3.4) is known explicitly. Multiplying both sides of (3.2) by the integrating factor $e^{\theta t}$ and integrating from t' to t yields

$$x_t = \mu + (x_{t'} - \mu)e^{-\theta(t-t')} + \nu \int_{t'}^{t} e^{-\theta(t-s)}\, dW_s. \tag{3.5}$$

The stochastic integral in (3.5) is a Gaussian random variable with zero mean and variance $\frac{\nu^2}{2\theta}(1 - e^{-2\theta(t-t')})$. Hence the conditional distribution is

$$x_t \,|\, x_{t'} \;\sim\; \mathcal{N}\big(\bar{x},\; \sigma_c^2\big)\,,$$

where

$$\bar{x} = \mu + (x_{t'} - \mu)\, e^{-\theta\Delta t},$$
$$\sigma_c^2 = \frac{\nu^2}{2\theta}\left(1 - e^{-2\theta\Delta t}\right),$$

and $\Delta t = t - t'$ is the time step. It is convenient to introduce the notation

$$\rho = e^{-\theta\Delta t}, \qquad \sigma^2 = \frac{\nu^2}{2\theta},$$

so that $\sigma_c^2 = \sigma^2(1-\rho^2)$ and $\bar{x} = \mu + \rho(x_{t'} - \mu)$. For a stationary OU process ($\theta > 0$), the initial distribution is given by the stationary (invariant) measure

$$x_0 \sim \mathcal{N}\big(\mu, \sigma^2\big)\,.$$

Introducing centered coordinates $z = x - \mu$, the transition density and the stationary density take the form

$$p(z_t|z_{t'}) = \frac{1}{\sigma_c\sqrt{2\pi}} \exp\left(-\frac{(z_t - \rho\, z_{t'})^2}{2\sigma_c^2}\right), \tag{3.6}$$

$$p_0(z) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{z^2}{2\sigma^2}\right). \tag{3.7}$$

3.3 Transfer matrix method for likelihood evaluation

Substituting the factorized joint density of the stationary OU process into (3.3), the likelihood function takes the form

$$L = \int \left[\prod_{t=0}^{T} f_t(x_t)\right] \cdot p_0(x_0) \prod_{t=1}^{T} p(x_t|x_{t-1})\, dx_0 \cdots dx_T,$$

where $f_t(x_t) = c(u_{1t}, u_{2t}; \Psi(x_t))$ is the copula density evaluated at the t-th observation for a given value of the latent process. The Markov property of the OU process ensures that this integral has a chain structure.

Backward messages. Define the backward messages recursively. Starting from the last observation, set

$$m_T(x_{T-1}) = \int f_T(x_T)\, p(x_T|x_{T-1})\, dx_T. \tag{3.8}$$

For $t = T-1, \ldots, 1$:

$$m_t(x_{t-1}) = \int f_t(x_t)\, p(x_t|x_{t-1})\, m_{t+1}(x_t)\, dx_t. \tag{3.9}$$

The likelihood is then

$$L = \int f_0(x_0)\, p_0(x_0)\, m_1(x_0)\, dx_0. \tag{3.10}$$

Each message m_t is a function of a single variable, so the problem is reduced from a $(T+1)$-dimensional integral to a sequence of T one-dimensional integrals.

Discretization on a grid. In centered coordinates $z = x - \mu$, introduce a uniform grid of K points

$$z_j = -R\sigma + (j-1)\Delta z, \quad j = 1, \ldots, K, \qquad \Delta z = \frac{2R\sigma}{K-1},$$

where $R > 0$ is a parameter controlling the grid extent in units of the stationary standard deviation σ. Each function is represented as a vector of length K:

$$\mathbf{f}_t = (f_t(z_1+\mu), \ldots, f_t(z_K+\mu)), \qquad \mathbf{m}_t = (m_t(z_1), \ldots, m_t(z_K)).$$

Transfer matrix. The transition density (3.6) is discretized as the matrix $\mathbf{T} \in \mathbb{R}^{K\times K}$:

$$T_{ji} = p(z_t = z_i \mid z_{t-1} = z_j) = \frac{1}{\sigma_c\sqrt{2\pi}} \exp\left(-\frac{(z_i - \rho\, z_j)^2}{2\sigma_c^2}\right),$$

where the row index j corresponds to the previous state $z_{t-1} = z_j$ and the column index i to the next state $z_t = z_i$.

The integrals in (3.8)–(3.9) are approximated by the trapezoidal quadrature. Let $\mathbf{w} \in \mathbb{R}^K$ be the vector of quadrature weights:

$$w_i = \Delta z \;\text{ for } i = 2, \ldots, K-1, \qquad w_1 = w_K = \frac{\Delta z}{2}.$$

Define the weighted transfer matrix $\widetilde{\mathbf{T}}$ with entries $\widetilde{T}_{ji} = T_{ji} \cdot w_i$. Then the backward recursion (3.9) becomes the matrix-vector operation

$$\mathbf{m}_t = \widetilde{\mathbf{T}}\,(\mathbf{f}_t \odot \mathbf{m}_{t+1}), \qquad t = T-1, \ldots, 1, \tag{3.11}$$

where $\odot$ denotes the Hadamard (element-wise) product, and the initial step is $\mathbf{m}_T = \widetilde{\mathbf{T}}\,\mathbf{f}_T$.

Final convolution. The stationary density (3.7) is discretized as the vector $\mathbf{p}_0 = (p_0(z_1), \ldots, p_0(z_K))$. The likelihood (3.10) is then

$$L \approx \sum_{i=1}^{K} f_0(z_i+\mu) \cdot p_0(z_i) \cdot m_1(z_i) \cdot w_i = \mathbf{w}^\top (\mathbf{f}_0 \odot \mathbf{p}_0 \odot \mathbf{m}_1).$$

Numerical stabilization. During the backward pass, the entries of $\mathbf{m}_t$ may decay (or grow) exponentially, leading to numerical underflow or overflow. To avoid this, after each step we normalize

$$\mathbf{m}_t \leftarrow \frac{\mathbf{m}_t}{c_t}, \qquad c_t = \max_j |m_t(z_j)|,$$

and accumulate the log-scale $S = \sum_t \log c_t$. The log-likelihood is then

$$\log L = \log\left(\sum_{i=1}^{K} f_0(z_i+\mu) \cdot p_0(z_i) \cdot m_1(z_i) \cdot w_i\right) + S. \tag{3.12}$$

Computational complexity. Each of the T backward steps requires one matrix-vector multiplication of cost $O(K^2)$ and one element-wise product of cost $O(K)$. The total complexity is $O(TK^2)$, which is efficient for typical values $T \sim 10^2$–10^3 and $K \sim 10^2$–10^3.

Comparison with Monte Carlo methods. The standard Monte Carlo estimator of the likelihood (the p-sampler (2.6)) computes $\hat{L} = \frac{1}{N}\sum_{i=1}^{N}\prod_t c(u_{1t}, u_{2t}; \Psi(x_t^{(i)}))$, then takes $\log \hat{L}$. By Jensen's inequality, $\mathbb{E}[\log \hat{L}] \leq \log L$, so this estimator is biased downward, with bias of order $O(1/N)$. Moreover, the variance of the product $\prod_t c(\cdot)$ grows exponentially with T, requiring an exponentially large number of trajectories for convergence. The transfer matrix method avoids both issues by computing the integral deterministically on a grid.

Finding copula parameters. To estimate the parameters $\alpha = (\theta, \mu, \nu)$ of the latent OU process, we follow the maximum likelihood approach.

Problem 1

$$\alpha = \arg\max_{\alpha'} \ \log L(\alpha'; U_1, U_2),$$

where the likelihood function (3.12) is considered.

3.4 Implementation details

We now discuss several implementation aspects of the transfer matrix method that are important for achieving both accuracy and computational efficiency, particularly for long time series.

Adaptive grid refinement. The accuracy of the trapezoidal quadrature in (3.11) depends on how well the grid resolves the transition kernel. The conditional standard deviation is $\sigma_c = \sigma\sqrt{1-\rho^2}$, where $\rho = e^{-\theta\Delta t}$, and the grid covers the range $[-R\sigma,\ R\sigma]$ of the stationary distribution. When $\theta\Delta t \ll 1$ (i.e., the observation frequency is high relative to the mean-reversion rate), we have $\rho \approx 1$ and $\sigma_c \approx \sigma\sqrt{2\theta\Delta t} \ll \sigma$: the transition kernel is much narrower than the stationary distribution. If the grid spacing $\Delta z = 2R\sigma/K$ is too coarse compared to σ_c, the Gaussian kernel $p(z_t|z_{t-1})$ is under-resolved and the quadrature error becomes significant. Conversely, when $\theta\Delta t \gg 1$, we have $\rho \approx 0$ and $\sigma_c \approx \sigma$: the kernel width is comparable to the grid range and a moderate K suffices.

To address this, we employ an adaptive rule for the grid size: given the user-specified minimum K and grid range R, we compute

$$K_{\text{eff}} = \max\left(K,\ \left\lceil \frac{2R\sigma}{\sigma_c/n_{\text{pts}}} \right\rceil + 1\right), \tag{3.13}$$

which guarantees at least n_{pts} grid points per conditional standard deviation σ_c (by default $n_{\text{pts}} = 4$). This ensures adequate resolution of the transition kernel regardless of the OU parameters. For fast mean-reversion relative to the time step ($\theta\Delta t \gg 1$), the kernel is wide and $K_{\text{eff}} \approx K$; for slow mean-reversion or high-frequency observations ($\theta\Delta t \ll 1$), the kernel becomes narrow and the grid is automatically refined.

We note that the integrand $f_t(z) = c(u_{1t}, u_{2t}; \Psi(z+\mu))$ in the quadrature (3.11) is not a simple unimodal function. For copula families where the link function Ψ maps both large positive and large negative values of z to large copula parameters (e.g., $\Psi(x) = x\tanh(x)$), the function f_t can exhibit a bimodal, approximately symmetric structure with two peaks on either side of $z = 0$. This shape arises because both z and $-z$ yield the same copula parameter $\Psi(z+\mu)$ when μ is small. The adaptive grid refinement ensures that both peaks, as well as the region between them, are adequately resolved.

Dense and sparse transfer matrices. When the transition kernel is wide (i.e., the bandwidth of the matrix $\widetilde{\mathbf{T}}$ covers a significant fraction of the grid), the transfer matrix is essentially full and the dense matrix-vector multiplication $\widetilde{\mathbf{T}}\mathbf{v}$ with cost $O(K^2)$ is optimal. However, for long time series with many observations, Δt is small, the correlation $\rho = e^{-\theta\Delta t}$ is close to 1, and the conditional variance $\sigma_c^2 = \sigma^2(1-\rho^2)$ becomes small. In this regime the Gaussian kernel is sharply peaked: the matrix $\widetilde{\mathbf{T}}$ has only $O(b)$ significant entries per row, where $b = \lceil 5\sigma_c/\Delta z \rceil$ is the half-bandwidth.

When $b < K/4$, we store $\widetilde{\mathbf{T}}$ as a sparse matrix in CSR (Compressed Sparse Row) format. The matrix-vector product then costs $O(K \cdot b)$ instead of $O(K^2)$, which can provide a substantial speedup for large K. The selection between dense and sparse representations is made automatically based on the ratio b/K:

$$\text{storage} = \begin{cases} \text{dense}, & \text{if } b \geq K/4, \\ \text{sparse (CSR)}, & \text{if } b < K/4. \end{cases}$$

This strategy is particularly beneficial for high-frequency data, where T can be on the order of 10^4 and the adaptive grid (3.13) may increase K to several thousand.

Forward pass. In addition to the backward pass used for likelihood evaluation, several quantities of interest—the smoothed copula parameter $\bar{\theta}_k = \mathbb{E}[\Psi(x_k)|U_{1,k-1}, U_{2,k-1}]$, the mixture Rosenblatt transform (4.1), and the predictive distribution of x_T—require a *forward pass*. In the forward pass, we propagate the density $\alpha_k(z) \propto p(z_k|U_{1,k-1}, U_{2,k-1})$ from $k = 0$ to $k = T$ by alternating updates (absorption of observation u_k) and predictions (propagation through the transition kernel):

$$\alpha_k(z) \leftarrow f_k(z) \cdot \alpha_k(z),$$
$$\alpha_{k+1}(z') = \int p(z'|z)\, \alpha_k(z)\, dz \approx \widetilde{\mathbf{T}}^\top (\boldsymbol{\alpha}_k \odot \mathbf{w}),$$

with normalization $\boldsymbol{\alpha}_k \leftarrow \boldsymbol{\alpha}_k / \max_j |\alpha_k(z_j)|$ to prevent under/overflow. Note that the forward pass uses the transpose $\widetilde{\mathbf{T}}^\top$, which for the sparse CSR format is computed via the CSC representation at no additional cost.

The *smoothed copula parameter* is computed as the predictive expectation

$$\bar{\theta}_k = \mathbb{E}\left[\Psi(x_k)|U_{1,k-1}, U_{2,k-1}\right] = \frac{\int\limits_{F_1 \times \ldots \times F_{t_k}} \Psi(x_k) \prod\limits_{i=1}^{t_{k-1}} c(u_{1i}, u_{2i}, \Psi(x_i)) d\nu_{t_1,\ldots,t_k}}{\int\limits_{F_1 \times \ldots \times F_{t_{k-1}}} \prod\limits_{i=1}^{t_{k-1}} c(u_{1i}, u_{2i}, \Psi(x_i)) d\nu_{t_1,\ldots,t_{k-1}}} \approx$$
$$\approx \sum_{j=1}^{K} \Psi(z_j + \mu) \cdot \hat{p}_k(z_j) \cdot w_j.$$

where $\hat{p}_k(z_j) = \alpha_k(z_j) / \sum_{i=1}^{K} \alpha_k(z_i)\, w_i$ are the normalized forward weights at step k. This quantity uses only data up to time $k-1$ and provides a one-step-ahead estimate of the copula parameter. Note that $\bar{\theta}_k$ is not used in likelihood evaluation (which is performed via the backward pass), but serves as a diagnostic tool for visualising the dynamics of the dependence parameter and for computing the mixture Rosenblatt transform described in Section 4.

All forward-pass quantities share the same grid, transfer operator, and propagation loop; they differ only in the per-step computation performed on the predictive weights. This allows a unified implementation with a callback mechanism, reducing code duplication and ensuring that the adaptive grid and sparse/dense selection are applied consistently across all transfer matrix operations.

3.5 Extension to vine copulas

The transfer matrix method developed above for bivariate copulas extends naturally to the vine copula framework described in Section 2.3. When combining the C-vine construction with the SCAR model, each edge copula is equipped with its own latent OU process with parameters $\alpha_e = (\theta_e, \mu_e, \nu_e)$. The transfer matrix likelihood and the forward pass are computed independently for each edge. This edge-by-edge estimation strategy, analogous to the sequential approach in [ACM16], allows the $d(d-1)/2$ bivariate SCAR models to be fitted separately, greatly reducing the computational burden. For edges at higher tree levels, where the conditional dependence is typically weak, some copula parameters may collapse to a constant (i.e., $\nu_e \approx 0$), effectively reducing to a static model.

In the vine construction, the conditional pseudo-observations $u_{a|S}$ at higher tree levels are computed via the h-function (2.3). When the copula parameter is time-varying and latent, a point estimate of θ_t would ignore the uncertainty of the latent process. Instead, we compute the *mixture h-function* by

integrating the h-function over the predictive distribution of the latent state, in complete analogy with the mixture Rosenblatt transform (4.1) described in Section 4:

$$\tilde{h}_k = \mathbb{E}\big[h(u_{2k}, u_{1k}; \Psi(x_k)) \,\big|\, U_{1,k-1}, U_{2,k-1}\big] \approx \sum_{j=1}^{K} h(u_{2k}, u_{1k}; \Psi(z_j + \mu)) \cdot \hat{p}_k(z_j) \cdot w_j,$$

where $\hat{p}_k$ denotes the normalized forward weights from the transfer matrix pass for the given edge. In fact, the mixture Rosenblatt transform used for the goodness-of-fit test and the mixture h-function used in the vine construction are the same computation; they differ only in their purpose (model validation vs. propagation of pseudo-observations to higher tree levels).

4 Goodness of fit for dynamic copula models

To assess the quality of copula models, we use an approach based on the Rosenblatt transform. Consider a d-dimensional copula $C(u_1, ..., u_d)$, a vector $U = (u_1, ..., u_d)$ and the conditional distribution functions

$$C(u_i|u_1, ..., u_{i-1}) = \frac{\partial^{i-1} C(u_1, ..., u_i, 1, ..., 1)/\partial u_1 ... \partial u_{i-1}}{\partial^{i-1} C(u_1, ..., u_{i-1}, 1, ..., 1)/\partial u_1 ... \partial u_{i-1}}, \quad i = 2, ..., d.$$

If the distribution parameters are chosen correctly [HH15], the quantities

$$\begin{aligned} u_1' &= u_1, \\ u_2' &= C(u_2|u_1), \\ u_3' &= C(u_3|u_1, u_2), \\ &\vdots \\ u_d' &= C(u_d|u_1, ..., u_{d-1}) \end{aligned}$$

and the corresponding vector $U' = (u_1', ..., u_d')$ belong to an independent copula, i.e., the random variables $u_1', ..., u_d'$ are pairwise independent and uniformly distributed on $[0, 1]$. The transformation $R : U \to U'$ is called the Rosenblatt transformation. The implementation details of this method are given in [HH15].

For the bivariate case ($d = 2$), the Rosenblatt transform reduces to

$$u_1' = u_1, \qquad u_2' = h(u_2, u_1; \theta) \coloneqq C(u_2|u_1; \theta) = \frac{\partial C(u_1, u_2; \theta)}{\partial u_1},$$

where h is the h-function of the copula.

In the case of a constant copula parameter (MLE model), θ is a fixed scalar. For the stochastic copula model, the copula parameter $\theta_t = \Psi(x_t)$ is time-dependent and latent. Rather than plugging in a point estimate of x_t, which would ignore the uncertainty of the latent process, we compute the *mixture Rosenblatt transform* by integrating the h-function over the predictive distribution of the latent state:

$$u_{2,t}' = \mathbb{E}\big[h(u_{2t}, u_{1t}; \Psi(x_t)) \,\big|\, U_{1,t-1}, U_{2,t-1}\big] = \int h(u_{2t}, u_{1t}; \Psi(x_t))\, p(x_t \,|\, U_{1,t-1}, U_{2,t-1})\, dx_t. \tag{4.1}$$

The predictive density $p(x_t \,|\, U_{1,t-1}, U_{2,t-1})$ is obtained from the transfer matrix forward pass described in Section 3.3. On the grid, the computation reduces to a weighted sum:

$$u_{2,t}' \approx \sum_{j=1}^{K} h(u_{2t}, u_{1t}; \Psi(z_j + \mu)) \cdot \hat{p}_t(z_j) \cdot w_j,$$

where $\hat{p}_t(z_j)$ is the normalized forward message at time t and w_j are the quadrature weights.

In our case, when the copula parameter is time-dependent, we consider ordered sets $U_t = (u_{1t}, ..., u_{dt})$ – the ordering being determined by the simple ordering of the data in the original set.

From the independence and uniform distribution of the vector U' it follows that the univariate random variable Y:

$$Y = F_{\chi^2_d}\left(\sum_{i=1}^{d} \Phi^{-1}(u'_i)^2\right)$$

is uniformly distributed on $[0,1]$, where $F_{\chi^2_d}$ is the distribution function of the χ^2-distribution with d degrees of freedom, Φ^{-1} is the inverse function to the distribution function of the standard normal distribution. The uniformity of Y can be checked, for example, by using the Cramér-von Mises criterion.

5 Numerical results

In this section we apply the proposed continuous SCAR model with the transfer matrix likelihood evaluation to real cryptocurrency market data. We consider two datasets of different lengths and several copula families to demonstrate that the method works consistently across different specifications. We also present a multivariate example using a C-vine copula, analyse convergence of the numerical scheme with respect to the grid size K, compare the computational performance of dense and sparse transfer matrices, and quantify the Jensen bias inherent in the Monte Carlo p-sampler.

5.1 Data description

We use two datasets of BTC-USD and ETH-USD returns.

Dataset 1 (daily, $T = 1460$). Daily log-returns of BTC-USD and ETH-USD from January 2, 2020 to December 31, 2023, computed as $r_t = \log(P_t/P_{t-1})$. This dataset represents a moderate-length time series typical in financial applications.

Dataset 2 (30-minute, $T = 12{,}000$). Intraday 30-minute log-returns of BTC and ETH from January 1, 2020 to September 7, 2020. This high-frequency dataset tests the scalability of the transfer matrix method and highlights the regime where sparse matrices and adaptive grid refinement become essential.

In both cases, log-returns are transformed to pseudo-observations via the rank transformation (2.2). This corresponds to a nonparametric estimation of marginal distributions, so that the copula model is fitted independently of any parametric assumption on the marginals (the so-called *inference functions for margins* approach [Joe97]).

For the multivariate experiment, we additionally use daily returns of BNB-USD, ADA-USD, XRP-USD, and DOGE-USD over a sub-period of 250 trading days (January 2 to September 7, 2020), yielding a 6-dimensional dataset.

5.2 Bivariate model comparison across copula families

To demonstrate that the proposed SCAR-TM-OU model provides consistent improvements regardless of the copula specification, we compare MLE (constant θ) with SCAR-TM-OU for four Archimedean copula families: Gumbel-180, Frank, Clayton, and Joe-180. Each model is assessed via the log-likelihood $\log L$, the AIC (with $k = 1$ parameter for MLE and $k = 3$ for SCAR-TM-OU), the BIC, and the Cramér–von Mises goodness-of-fit p-value described in Section 4.

Dataset 1: daily returns ($T = 1460$). The results are presented in Table 2. Across all four copula families, the SCAR-TM-OU model yields a substantially higher log-likelihood than the constant-parameter MLE. The improvement ranges from approximately 77 nats (Joe-180) to 103 nats (Frank). The AIC and BIC values confirm that this improvement is not an artefact of the additional parameters: the ΔAIC (defined as $\mathrm{AIC}_{\mathrm{MLE}} - \mathrm{AIC}_{\mathrm{SCAR}}$) ranges from 150 to 202, vastly exceeding the standard threshold of 10 for decisive evidence.

Regarding model adequacy, the MLE model is rejected by the GoF test at the 5% level for Clayton and Joe-180 ($p < 0.001$), and at the 1% level for Gumbel-180 ($p = 0.009$). Frank is borderline ($p = 0.049$). In contrast, the SCAR-TM-OU model passes the GoF test for Gumbel-180 ($p = 0.769$), Clayton ($p = 0.212$), and Joe-180 ($p = 0.259$). For Frank the p-value is 0.051, which is marginal but substantially improved over the MLE fit.

Table 2: Bivariate model comparison on daily BTC–ETH returns ($T = 1460$). For MLE, $k = 1$ parameter; for SCAR-TM-OU, $k = 3$ parameters $(\theta_{\mathrm{OU}}, \mu, \nu)$.

Copula	Method	$\log L$	AIC	BIC	GoF p
Gumbel-180	MLE	955.63	−1909.26	−1903.97	0.009
	SCAR-TM	1045.50	−2085.00	−2069.14	0.769
Frank	MLE	843.02	−1684.04	−1678.75	0.049
	SCAR-TM	945.83	−1885.66	−1869.80	0.051
Clayton	MLE	873.86	−1745.73	−1740.44	< 0.001
	SCAR-TM	956.10	−1906.20	−1890.34	0.212
Joe-180	MLE	871.63	−1741.26	−1735.97	< 0.001
	SCAR-TM	948.88	−1891.76	−1875.91	0.259

Dataset 2: 30-minute returns ($T = 12{,}000$). Table 3 shows the results for the high-frequency dataset. Here the discrepancy between the static and dynamic models is even more pronounced. The constant-parameter MLE is rejected at any reasonable significance level for all four families ($p < 0.001$). The SCAR-TM-OU model improves the log-likelihood by 543–581 nats, with Gumbel-180 achieving the highest log-likelihood (5075.0) and the only GoF p-value above 0.05 ($p = 0.432$). The remaining three families show improved but still low p-values under the SCAR model, suggesting that even the time-varying specification may not fully capture the complex intraday dependence dynamics for these particular copula forms.

Table 3: Bivariate model comparison on 30-minute BTC–ETH returns ($T = 12{,}000$).

Copula	Method	$\log L$	AIC	BIC	GoF p
Gumbel-180	MLE	4525.48	−9048.96	−9041.58	< 0.001
	SCAR-TM	5075.00	−10143.99	−10121.84	0.432
Frank	MLE	4266.48	−8530.96	−8523.58	< 0.001
	SCAR-TM	4689.61	−9373.21	−9351.06	0.009
Clayton	MLE	3857.96	−7713.92	−7706.54	< 0.001
	SCAR-TM	4439.05	−8872.10	−8849.95	< 0.001
Joe-180	MLE	3787.45	−7572.91	−7565.53	< 0.001
	SCAR-TM	4330.88	−8655.76	−8633.61	< 0.001

For $T = 12{,}000$ and $\Delta t \approx 8.3 \times 10^{-5}$, the conditional standard deviation σ_c is very small relative to the stationary σ, so the adaptive grid rule (3.13) increases K well above the default value and the sparse transfer matrix representation is selected automatically. This makes the computation tractable even for this large dataset.

Figure 1 summarises the AIC improvement and GoF p-values for Dataset 1 across all four copula families.

5.3 Smoothed copula parameter

A key advantage of the stochastic model is its ability to capture the time-varying dependence structure. Figure 2 shows the predictive estimate $\bar{\theta}_k = \mathbb{E}[\Psi(x_k) \mid U_{1,k-1}, U_{2,k-1}]$ for the Gumbel-180 copula on Dataset 1, together with its 30-day rolling mean and the constant MLE estimate.

The Gumbel-180 parameter θ is directly related to the strength of lower-tail dependence (joint drawdowns). The time-varying estimate reveals several episodes of elevated dependence that coincide with known market events: the COVID-19 crash in March 2020, the period of heightened market stress in mid-2021, the prolonged "crypto winter" of mid-2022, and the FTX collapse in November 2022. Conversely, the parameter decreases during calmer periods (e.g., 2021 Q3–Q4, 2023 H1). The constant MLE estimate ($\hat{\theta} = 2.83$) averages over these regimes and, as shown in Table 2, yields a poor fit.

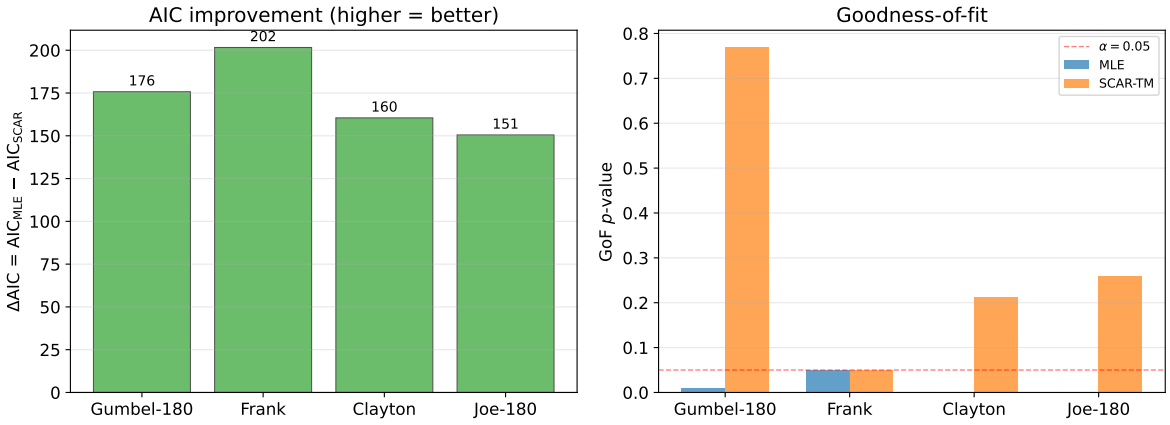

Figure 1: Left: AIC difference $\Delta\text{AIC} = \text{AIC}_{\text{MLE}} - \text{AIC}_{\text{SCAR}}$ (positive values favour the stochastic model). Right: GoF p-values; the dashed line indicates the 5% significance level. Dataset 1, daily BTC–ETH, $T = 1460$.

The estimated OU parameters are $\hat{\theta}_{\text{OU}} = 58.96$, $\hat{\mu} = 1.488$, $\hat{\nu} = 4.531$, giving a characteristic mean-reversion time of $1/\hat{\theta}_{\text{OU}} \approx 6$ days. The smoothed parameter ranges from 1.57 to 6.45 with a mean of 3.38.

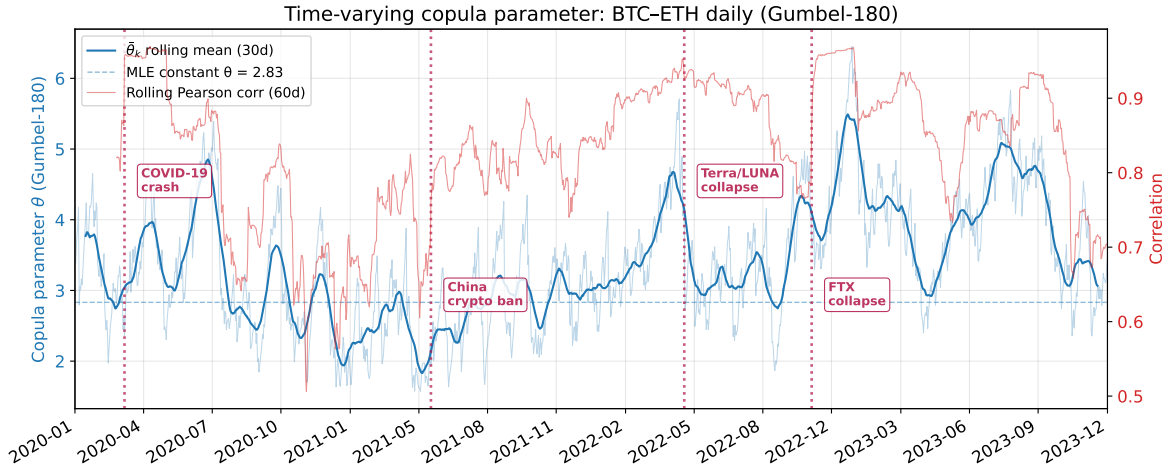

Figure 2: Time-varying Gumbel-180 copula parameter for daily BTC–ETH returns ($T = 1460$). Light blue: raw predictive estimate $\bar{\theta}_k$; bold blue: 30-day rolling mean; dashed: constant MLE estimate.

5.4 Multivariate example: C-vine copula

To demonstrate that the proposed method extends naturally to higher dimensions, we fit a C-vine copula to the 6-dimensional cryptocurrency dataset ($d = 6$, $T = 250$). The C-vine structure is estimated automatically by the algorithm, which selects the copula family and rotation for each edge via AIC. We compare four models: C-vine with SCAR-TM-OU, C-vine with MLE, Student-t copula, and Gaussian copula. The results are shown in Table 4.

The C-vine with SCAR-TM-OU achieves the highest log-likelihood and comfortably passes the GoF test ($p = 0.789$). The C-vine with MLE also performs well ($p = 0.210$), while the elliptical copulas (Student-t and Gaussian) are decisively rejected. This illustrates the value of both the vine structure and the time-varying parameter specification in higher dimensions.

The fitted C-vine structure uses Gumbel-180 copulas on most edges in the first two trees (capturing lower-tail dependence), with Frank, Gaussian, and Joe copulas appearing in the higher trees where conditional dependencies are weaker. Full details of the vine structure are available in the accompanying

Table 4: Multivariate model comparison on 6-dimensional cryptocurrency data ($d = 6$, $T = 250$).

Model	$\log L$	GoF statistic	GoF p-value
C-vine SCAR-TM	908.69	0.064	0.789
C-vine MLE	869.23	0.234	0.210
Student-t	764.42	1.585	< 0.001
Gaussian	761.00	2.553	< 0.001

code repository.

5.5 Convergence analysis with respect to grid size K

A central implementation parameter of the transfer matrix method is the grid size K. To study how the log-likelihood converges as K increases, we evaluate the transfer matrix log-likelihood at fixed parameters (Clayton copula, Dataset 1, with $\theta_{\text{OU}} = 0.01$, $\mu = 2.868$, $\nu = 1.0$, chosen to produce a wide stationary distribution where convergence is slowest). These parameters give a stationary standard deviation $\sigma = 7.07$ and a conditional standard deviation $\sigma_c = 0.026$, so the adaptive rule at 4 points per σ_c prescribes $K_{\text{eff}} \approx 10{,}800$.

Figure 3 (left panel) shows $\log L$ as a function of K. The log-likelihood is far from its converged value at $K = 200$ and stabilises to full numerical precision around $K \approx 2{,}000$–$3{,}000$, corresponding to approximately 1–2 points per σ_c. The adaptive rule at 4 points per σ_c is therefore conservative: it guarantees accuracy for the trapezoidal quadrature applied to an arbitrary integrand, but the smoothness of both the Gaussian transition kernel and the copula density allows convergence at a lower resolution.

The right panel of Figure 3 shows the wall-clock time, which scales linearly in K for the sparse transfer matrix implementation: from 0.04 s at $K = 20$ to approximately 1.0 s at $K = 7{,}000$ ($T = 1460$).

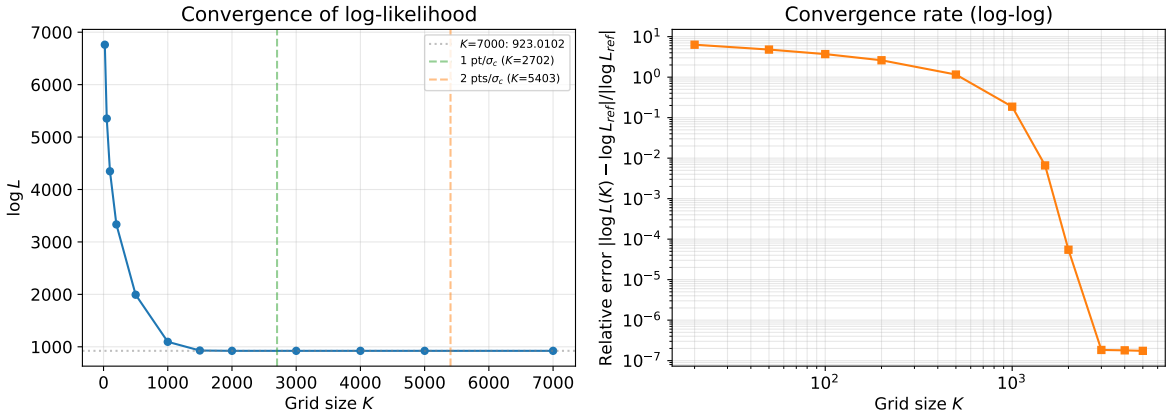

Figure 3: Convergence of the transfer matrix log-likelihood with respect to grid size K (Clayton copula, Dataset 1). Left: $\log L$ stabilises around $K \approx 2{,}000$–$3{,}000$, well below the adaptive rule's prescription $K_{\text{eff}} \approx 10{,}800$. Right: wall-clock time scales linearly with K.

5.6 Dense vs. sparse transfer matrices

When the time step Δt is small (e.g., high-frequency data), the OU transition kernel is narrow relative to the grid spacing, and the transfer matrix $\mathbf{M}$ becomes banded. In this regime, storing and multiplying $\mathbf{M}$ as a sparse (CSR) matrix reduces the per-step cost from $O(K^2)$ to $O(Kb)$, where b is the bandwidth.

Figure 4 compares the wall-clock time for dense and sparse matrix-vector products as a function of K, with fixed $T = 5{,}000$ (30-minute data, Gumbel-180 copula). At $K = 300$, dense evaluation takes approximately 5.1 s while sparse takes 1.5 s (a 3.4× speedup). As K increases, the advantage of sparse

storage grows: at $K = 5{,}000$, sparse is 7.4× faster (4.6 s vs. 34.1 s). The dense method exhibits $O(K^2)$ scaling, while the sparse method scales approximately linearly in K as expected.

We note that the dense matrix-vector product uses multi-threaded BLAS routines and saturates all CPU cores, while the sparse CSR matvec runs on a single thread. The reported wall-clock speedup thus *underestimates* the single-core advantage of the sparse method.

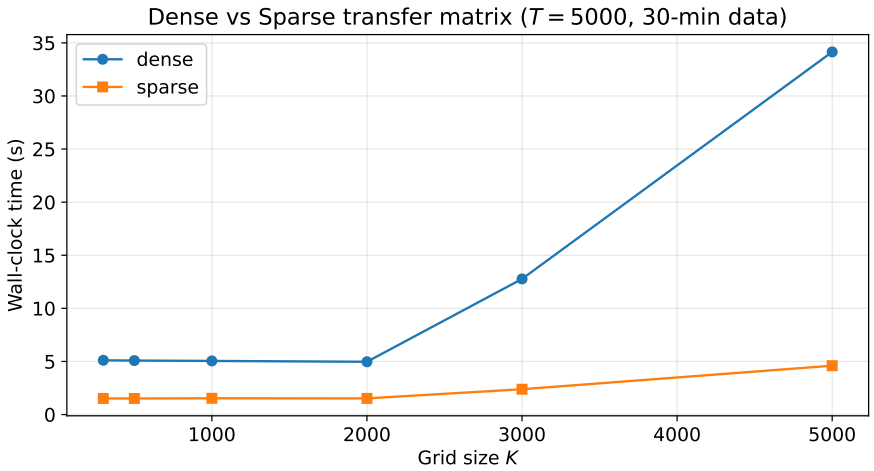

Figure 4: Wall-clock time for dense vs. sparse transfer matrix evaluation as a function of grid size K ($T = 5{,}000$, Gumbel-180, 30-minute data).

5.7 Jensen bias in Monte Carlo estimation

To demonstrate the bias inherent in the Monte Carlo p-sampler (2.6), we evaluate the log-likelihood at the optimal SCAR-TM-OU parameters for Gumbel-180 on Dataset 1 using $N_{\text{tr}} \in \{100, 500, 1000, 10{,}000, 100{,}000\}$ trajectories, with 50 independent replications for each N_{tr}. The transfer matrix value $\log L = 1045.50$ serves as the reference. The results are summarised in Table 5 and Figure 5.

Table 5: Monte Carlo p-sampler: Jensen bias and variance. True value: $\log L = 1045.50$ (TM). Dataset 1, Gumbel-180.

N_{tr}	mean $\log L$	std	bias
100	912.3	18.7	133.2
500	930.7	14.4	114.8
1,000	936.5	9.9	109.0
10,000	955.0	7.4	90.5
100,000	967.7	6.9	77.8

Even with $N_{\text{tr}} = 100{,}000$, the p-sampler underestimates the log-likelihood by approximately 78 nats, and the empirical bias decays as $O(N^{-0.08})$, far slower than the theoretical $O(1/N)$ rate.

This slow decay is a consequence of weight degeneracy: the product $\prod_{t=1}^{T} c(u_{1t}, u_{2t}; \Psi(x_t^{(i)}))$ has variance that grows exponentially with T, so only a tiny fraction of the N sampled trajectories carry non-negligible weight. The effective sample size $N_{\text{eff}} \ll N$, and the pre-factor $\exp(\gamma T)/N$ in the bias expression makes the decay appear nearly flat across computationally feasible values of N. The transfer matrix method avoids this problem entirely by computing the integral deterministically on a grid.

6 Conclusion

In this paper, we propose a stochastic copula model in continuous time for modeling time-varying dependence structures in financial data. Unlike existing SCAR models that rely on Monte Carlo inte-

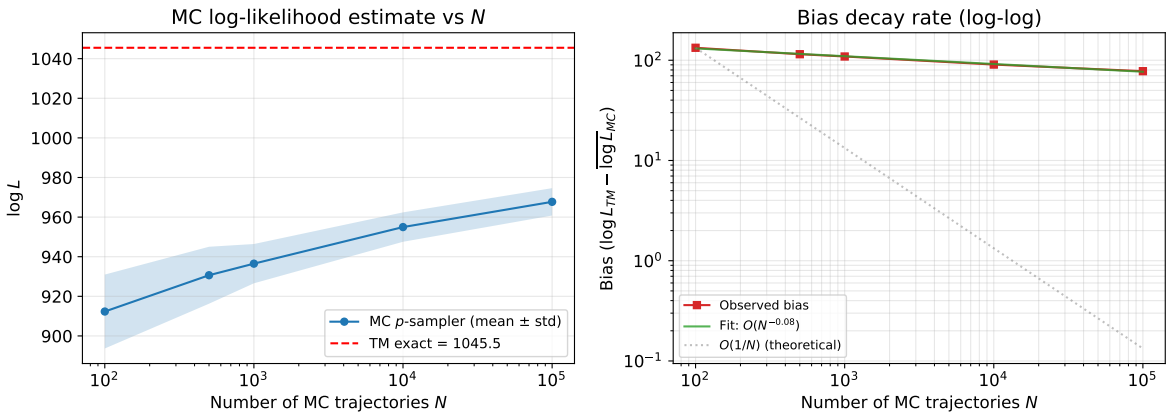

Figure 5: Left: mean log-likelihood from the MC p-sampler (with ± 1 std band) vs. the TM exact value (dashed red). Right: bias on a log-log scale with fitted decay exponent $\beta = 0.08$.

gration, we develop a transfer matrix method that exploits the Markov structure and known Gaussian transition density of the Ornstein-Uhlenbeck process to evaluate the likelihood function deterministically on a grid. This approach reduces the original high-dimensional integral to a sequence of one-dimensional integrals, each computed via matrix-vector multiplication with trapezoidal quadrature. The resulting scheme has computational complexity $O(TK^2)$ (or $O(TKb)$ with sparse matrices) and yields a numerically exact likelihood evaluation, free from the variance and downward bias inherent in the log-mean-exp Monte Carlo estimator.

We discuss implementation details that are essential for practical applicability: adaptive grid refinement that guarantees adequate resolution of the transition kernel regardless of the OU parameters, and the use of sparse transfer matrices that reduce the per-step cost from $O(K^2)$ to $O(Kb)$ when the kernel bandwidth b is small relative to K. These techniques make the method scalable to high-frequency datasets with T on the order of 10^4.

The proposed model was validated on cryptocurrency market data at two temporal resolutions: daily ($T = 1{,}460$) and 30-minute ($T = 12{,}000$). In the bivariate setting, we demonstrated consistent and substantial improvements across four Archimedean copula families (Gumbel-180, Frank, Clayton, Joe-180): the SCAR-TM-OU model improved the log-likelihood by 77–103 nats on daily data and 543–581 nats on 30-minute data, with AIC differences ranging from 150 to 1,095 in favour of the stochastic model. The goodness-of-fit test based on the mixture Rosenblatt transform confirmed model adequacy in most cases.

In the multivariate setting, a C-vine copula with SCAR-TM-OU estimation on 6-dimensional cryptocurrency data ($d = 6$, $T = 250$) achieved a log-likelihood of 908.7 and a GoF p-value of 0.789, substantially outperforming both the C-vine with constant parameters and the elliptical (Student-t and Gaussian) copulas. The smoothed copula parameter revealed economically interpretable dynamics, with elevated lower-tail dependence during known episodes of market stress.

The convergence analysis showed that the log-likelihood stabilises at approximately 2–3 points per conditional standard deviation σ_c, well below the conservative adaptive rule, making the method efficient in practice. The Monte Carlo p-sampler was shown to exhibit substantial downward bias even with 10^5 trajectories (approximately 78 nats for $T = 1{,}460$), with an empirical decay rate of $O(N^{-0.08})$, confirming the practical necessity of the deterministic transfer matrix approach.

As possible future research directions, we note investigating alternative stochastic processes (fractional Brownian motion or Lévy processes), extending the model to higher-dimensional copulas via more flexible vine constructions (R-vines), and applying the proposed framework to portfolio risk assessment and dynamic hedging.

The implementation of the proposed model is available as an open-source Python package at https://github.com/AANovokhatskiy/pyscarcopula.

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